

“Architectural Kata 2026”

A stab at architectural thinking in the age of AI

‘Tough but Fair’ Team

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1 Process

This is a KATA. It's not a real project. Hence, by definition there'll be (unrealistic) simplifications and a smoothing out of the problem space. The sharp edges of reality are extracted so we can focus on the crux of the topic or the areas we want to practice and improve on. This doesn't mean it can't be useful, or interesting!

The topics I personally want to exercise in this Kata are:

- ADRs: to provide a record and direction
- Architectural Characteristics to drive the process
- Using Domain Driven Design for analysis
- Using AI to help the architectural design
- Think about where AI integrates into an architecture
- Use Scrivener and Quarto for documentation

2 Architectural Characteristics

Keeping NFRs and architectural characteristics in mind orients the project on its real goals. It would be an early discussion topic, but these are some that are likely to be selected:

- **Extensibility:** As is typical there's a lot of unknowns in the future direction of the project. Meeting current requirements is table stakes; we need to be able to accommodate any and all future directions.
- **Reliability and Availability:** As a system with life-support and e-commerce aspects, at least it should degrade gracefully
- **Scalability:** There's a plan for the system to x3 in three years. This is equivalent to an annual increase of 44% per year. In ten years that's nearly 200x, but of course, diminishing returns and market saturation is probable before then, but this is worth planning for success.

Other characteristics that should be discussed:

- **Resiliency:** Various elements are likely to fail; the system should not persist
- **Cost and efficiency:** The estate is likely to discover new customer needs and will want to cater to them. These experiments shouldn't be too costly.
- **Usability:** The estate staff are not technical specialists so easy to learn and use will be a valuable aspect.
- **Security:** *Sine qua non*; but still worth making an explicit discussion about reputation and financial damage; and costs of best practices.

Year 3: 15,000 Year 4: 21,633
Year 5: 31,201 Year 6: 45,000
Year 7: 64,901 Year 8: 93,603
Year 9: 135,000 Year
10: 194,703

Of course, there are many other characteristics; discussion and sharing priorities with stakeholder would provide a keystone to strategic and technical direction.

3 Review of the Project Scope

The Project involves thinking about an estate that's seeking a new direction and is looking to a technological solutions to help them plot their new direction.
Event Page

3.1 Initial discussion of relevant details

Von Digitalis Estate is made up of:

- Currently: 5000 visitors a day: That's about equal to about a ten visitors a second.
 - This will generate a spike of traffic on open (and perhaps closing time) and a steady stream of telemetry and monitoring throughout the day
- Growth plans to 15,000 in three years: These are probably wild guesstimates: it would be wise to be able to scale and cope with a good deal more: Retaining the ability for x10 capacity increase would be wise. And given the uncertainty, new projects will likely to also be accommodated.
- There's several broad themes in several domains:
 - Tickets and passes
 - Visitor activity tracking and sensors: for trends and strategic planning
 - Asset telemetry: track activities, health and behaviour of the 40 amusement rides & 200 animals/plants/fish enclosures

'Predictions are hard, especially about the future' Niels Bohr

4 ADRs

4.1 0: Process and tools

Status: Accepted **Context:** Firstly let's set the tools, rules, and processes. ADRs, DDD and AI... **Decision:**

- Will proceed using ADRs
- Will use Domain Driven Design as a analysis framework
- Will seek AI feedback on each ADR: in a dedicated section
- Will use Scrivener to manage documents, links, research and deliverables.

Considered Options:

- Having a structured Approach is somewhat necessary

Consequences & Trade-offs:

- Positive: This will provide a record of decisions and be methodical
- Negative: Requires discipline.

4.2 1: Using DDD

Status: Accepted **Context:** Propose Domain-Driven Design as the analysis framework; we know there's complex business logic and need to decompose the monolith into services and have a start at classifying them.

4.3 Decision

There are six bounded contexts 1. **Ticketing & Passes** — sells tickets and family passes, issues entitlements, validates entry at the gate. 2. **Identity & Visitor Accounts** — visitor profiles, family/group membership, consent and privacy flags 3. **Payments** — thin ACL wrapper around a third-party provider; isolates card/PCI concerns. 4. **Ride Telemetry & Safety** — ingests sensor data (usage, faults, e-stop events) from the 40 rides. 5. **Animal Welfare Telemetry** — ingests health, feeding and population data across 55 enclosures. 6. **Visitor Insight & Personalization (Core)** — consumes events from every other context; runs demand prediction, anomaly detection, recommendations; publishes insight back out.

4 ADRs

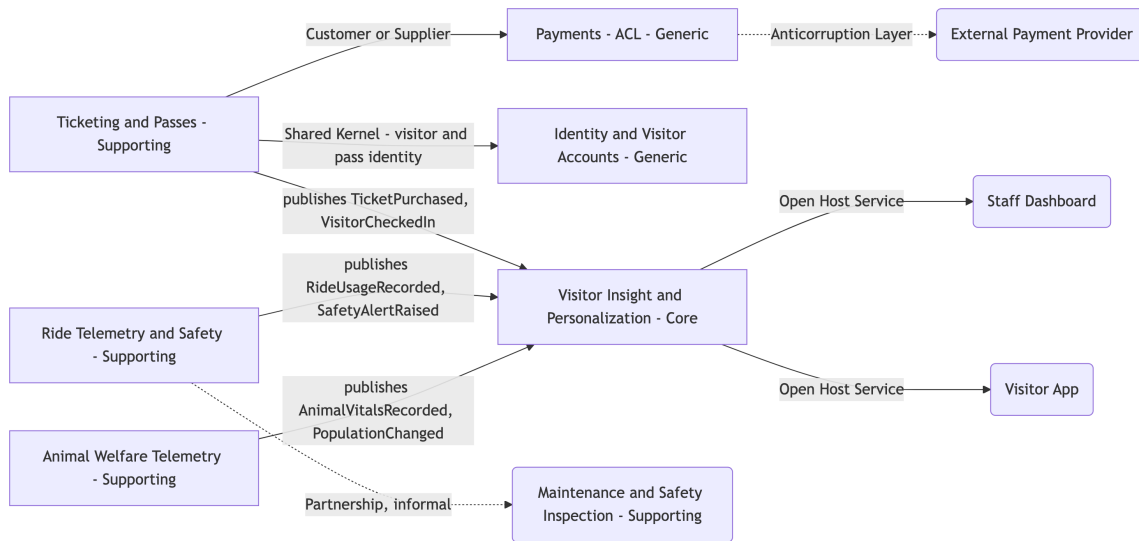


Figure 4.1

4.4 Decision-1

Every context already publishes domain events for integration. The remaining question is whether a context's own source of truth is the current state (state-oriented / CRUD) or the full append-only event log (event-sourced), with current state and every read model rebuilt from it. We apply event sourcing selectively: to the four contexts where an audit trail, incident reconstruction, or retroactive AI feature extraction has real business value — Ticketing & Passes, Ride Telemetry & Safety, Animal Welfare Telemetry, and Visitor Insight & Personalization. Identity & Visitor Accounts and Payments stay state-oriented — they're generic, solved-problem subdomains where an event store buys complexity without buying much. Fig 5,6,7 An append-only log and a visitor's right to be forgotten are in tension. Ticketing events will carry personal data — see the AI Feedback watch points below, and forthcoming ADR-8. Considered Options Two separate axes: A-C decide how the platform splits into contexts; D-F decide how those contexts keep their own state. A. Single monolithic "Estate Platform" with one shared data model — rejected: couples safety-critical ride data to marketing personalization, and blocks independent iteration on Visitor Insight without redeploying ticketing. B. Six bounded contexts, event-integrated — CHOSEN. Matches the three domains named in the brief, isolates the safety-critical ride path from analytics, and lets the AI-heavy core domain evolve independently. C. Two contexts only, Commerce vs Telemetry — rejected: still conflates ride safety with animal welfare (different cadence and criticality) and conflates ticketing with AI personalization (different rate of change). D. State-oriented (CRUD) persistence everywhere — rejected as the sole approach: throws away the audit trail and replay capability that Ride Telemetry, Animal Welfare, Ticketing and Visitor Insight would all benefit from. E. Event sourcing for every context, uniformly — rejected: forces an event store, snapshotting and projection-rebuild machinery onto Identity and Payments, where nothing about the problem calls for it. F. Event sourcing where it earns its keep — CHOSEN. Event-sourced for Ticketing & Passes,

Ride Telemetry & Safety, Animal Welfare Telemetry and Visitor Insight & Personalization; state-oriented for Identity and Payments. Consequences & Trade-offs Positive Safety-critical ride telemetry is isolated from experimental AI/personalization code — a bad model in Visitor Insight can't block gate entry or an e-stop signal. The core domain can iterate quickly and stay AI-heavy, decoupled from the more conservatively-changing supporting contexts. Matches the three domains already named in the brief, so it should translate cleanly into team and AI-assisted ownership. Event sourcing on Ticketing, Ride Telemetry, Animal Welfare and Visitor Insight gives a full audit trail, safety-incident reconstruction, and the ability to backfill new AI metrics from history instead of re-instrumenting the park. Negative Six contexts means six things to deploy, monitor and secure instead of one — real overhead a kata-scale build must account for (ADR-3). The Ticketing<->Identity Shared Kernel is a known coupling risk without a named owner. The six-way map needs discipline, or Visitor Insight will end up reaching directly into another context's database as a shortcut. Event-sourced contexts need snapshotting once their logs grow, and a schema-evolution (upcasting) strategy for event payloads over the estate's lifetime. An immutable event log makes deleting one visitor's personal data harder. AI Feedback (per ADR-0) The Shared Kernel between Ticketing & Passes and Identity & Visitor Accounts is the riskiest relationship on this map. Name a single owner for it now, before it calcifies. "Safety-critical" for Ride Telemetry & Safety is asserted here but not yet backed by a concrete latency/availability target — ADR-6 should quantify that so this isolation is testable, not aspirational. Collapsing every AI use case into one Visitor Insight context is a reasonable v1 simplification, but an animal-health anomaly detector and a marketing recommendation carry very different risk profiles in reality. Flag as a candidate future split, not settled. Open question for the Countess/stakeholder: is Payments truly generic, or does family-pass pricing/discounting belong closer to Ticketing than to a generic payment gateway? Worth confirming before ADR-5. Event sourcing on Ticketing & Passes means the log will contain personal data tied to an identifiable visitor. An append-only store and a GDPR erasure request are in direct tension — resolve with crypto-shredding (encrypt per-visitor, delete the key) or by keeping personal data out of the event payload and referencing it by pseudonymous ID instead. Treat as a hard requirement for ADR-8, not an afterthought. Snapshotting policy for Ride Telemetry & Safety and Animal Welfare Telemetry isn't decided here — with 40 rides and 55 enclosures streaming continuously, replaying from event zero stops being cheap well before year three's 15,000-visitor target. Related & anticipated Builds on ADR-0 (Process and tools). Anticipated follow-on ADRs: ADR-2 (edge-to-cloud connectivity), ADR-3 (event backbone), ADR-5 (ticketing and passes), ADR-6 (ride safety isolation), ADR-7 (elasticity), ADR-8 (visitor data privacy), ADR-9 (AI integration points).

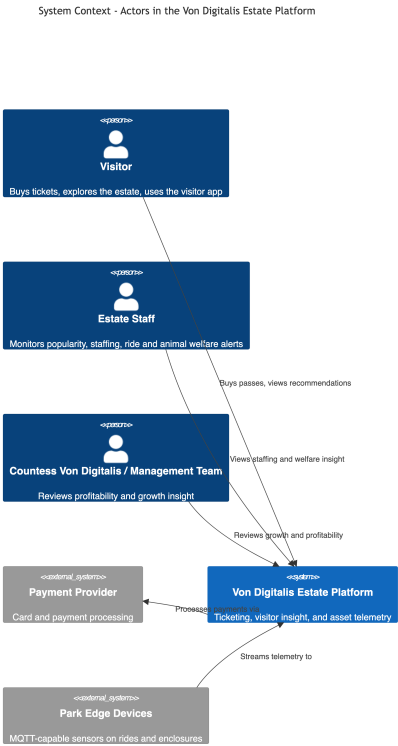


Figure 4.2

4 ADRs

Container Diagram - Von Digitalis Estate Platform

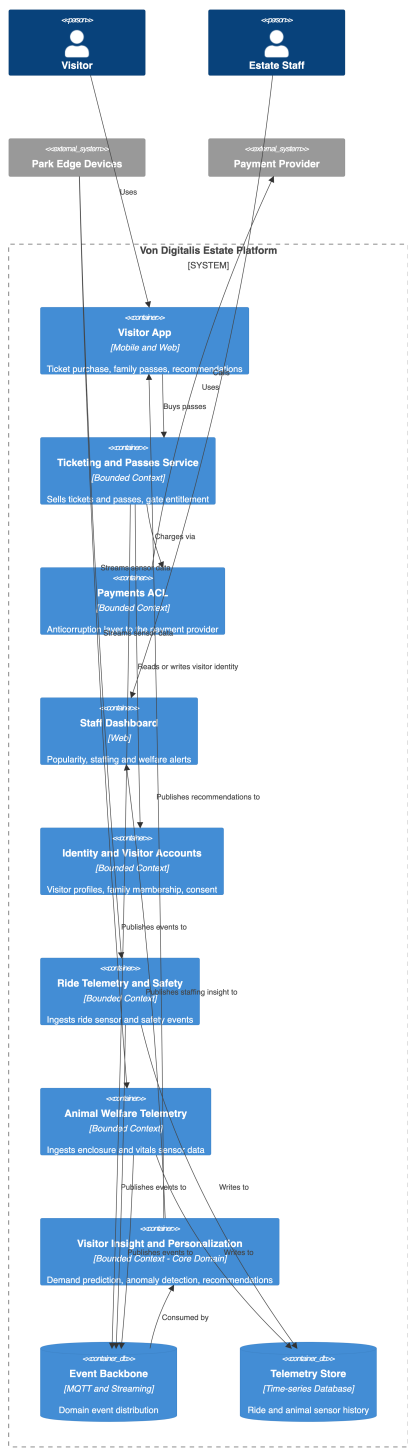


Figure 4.3

5 Deliverables

Summary of the Solution

- C4 based Mermaid Diagrams of the Domain Models & components
- ADRs
- Implementation Details
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